

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include "commonType.h"
4 #include "fileLoadWrite.h"
5
6 void loadData(node *head, node *tail){
7     FILE *fp;
8     elem *oldData=NULL, *newData=NULL;
9     const char *f = "file.txt";
10    if((fp=fopen(f,"r"))==NULL ){
11        printf("load file not Exist\n");
12        return;
13    }
14    newData = (elem *)malloc(sizeof(elem));
15    head->next = newData;
16    newData->prev = (elem *)head;
17
18    while(fscanf(fp, "%d %s %d", &newData->num
19        , newData->name, &newData->age) != EOF) {
20        head->cnt++;
21        tail->cnt++;
22        oldData = newData;
23        newData = (elem *)malloc(sizeof(elem));
24        oldData->next = newData;
25        newData->prev = oldData;
26    }
27    if(head->cnt == 0){
28        head->next = (elem*)tail;
29    }else{
30        oldData->next = (elem *)tail;
31        tail->prev = oldData;
32    }
33    free(newData);
34    fclose(fp);
35 }
36
37 void saveData(node *head, node *tail){
38     FILE *fp;
39     elem *list;
40     const char *f = "file.txt";
41     if((fp = fopen(f, "wt")) == NULL){
42         printf("file write error\n");
43         return;
44     }
45     list = head->next;
46     while(list != (elem *)tail){
47         fprintf(fp, "%d %s %d\n", list->num, list->name, list->age);
48         list = list->next;
49     }
50     fclose(fp);
51 }
52
```